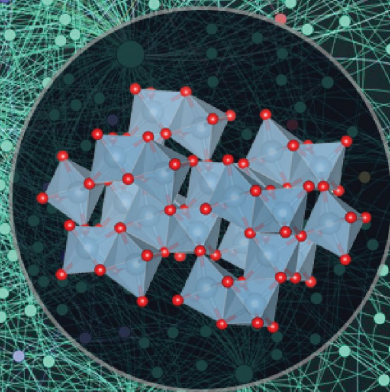
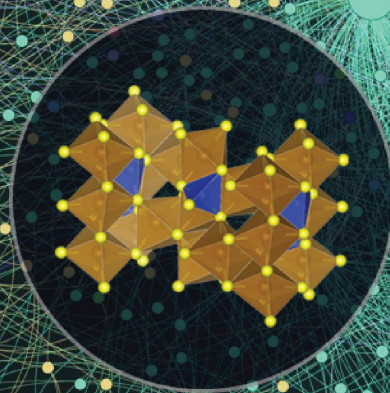


materials informatics

best practices



We recently published the definitive best practices guide for ML in MSE

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Machine Learning for Materials Scientists: An Introductory Guide toward Best Practices

Anthony Yu-Tung Wang, Ryan J. Murdock, Steven K. Kauwe, Anton O. Oliynyk, Aleksander Gurlo, Jakoah Brgoch, Kristin A. Persson, and Taylor D. Sparks*

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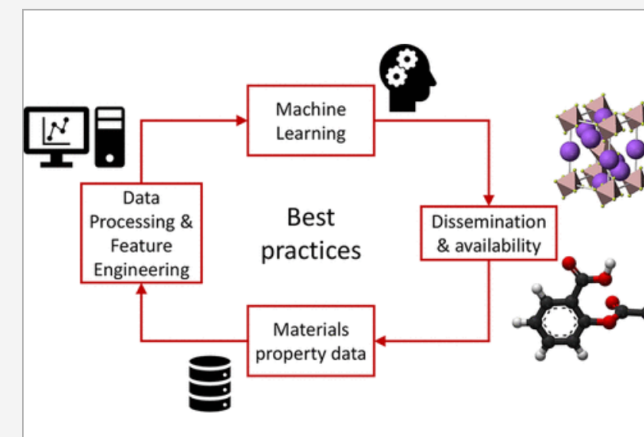
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Chemistry of Materials

Abstract

This Methods/Protocols article is intended for materials scientists interested in performing machine learning-centered research. We cover broad guidelines and best practices regarding the obtaining and treatment of data, feature engineering, model training, validation, evaluation and comparison, popular repositories for materials data and benchmarking data sets, model and architecture sharing, and finally publication. In addition, we include interactive Jupyter notebooks with example Python code to demonstrate some of the concepts, workflows, and best practices discussed. Overall, the data-driven methods and machine learning workflows and considerations are presented in a simple way, allowing interested readers to more intelligently guide their machine learning research using the suggested references, best practices, and their own materials domain expertise.



Jupyter notebooks are up on GitHub and in our worked examples folder

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 **anthony-wang** Merge pull request #8 from stefanbringuier/master 7b03d4f on Feb 18, 2021 15 commits

data	Pluto notebooks 1-3 completed. Some testing provided to ensure reaso...	13 months ago
notebooks	Improve inputting of missing values with column medians	14 months ago
pluto_notebooks	Pluto notebooks 1-3 completed. Some testing provided to ensure reaso...	13 months ago
.gitignore	Initial commit	2 years ago
LICENSE.md	Initial commit	2 years ago
README.md	Pluto notebooks 1-3 completed. Some testing provided to ensure reaso...	13 months ago
conda-env.yml	Update pandas-profiling package required version	14 months ago

README.md

BestPractices

Things that you should (and should *not*) do in your Materials Informatics research.

This is a repository containing the relevant Python code and Jupyter notebooks to the publication "Machine Learning for Materials Scientists: An Introductory Guide toward Best Practices".

These notebooks are included to illustrate a hypothetical Machine Learning project in materials science created following best practices. The goal of this project is to predict the heat capacity of materials given a chemical composition and condition (the measurement temperature).

About

Things that you should (and should not) do in your Materials Informatics research.

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